TEAM NEXORA · DOSM DATATHON 2026 · PROJECT GUIDE

# Understanding Our Whole Project

Geopolitical Risk and International Tourism Demand in Malaysia: A Machine Learning Forecasting Framework for 20 Source Markets. Chin William · Tang Xin Yi · Janis Ong Jing Yao · Tee Xin Yee · Advisor: Dr. Kelwin Tan Seen Tiong, UTAR

Each of us worked on one part. This guide explains the whole project so everyone understands all of it. It follows the order of our report (sections 1.0 to 6.0), adds what is in our project folder, and uses simple words. Every number comes from the report or the project's results.

> **The project in short.** We built a monthly dataset of tourist arrivals from Malaysia's 20 main source countries, together with exchange rates, geopolitical risk, oil prices and Malaysian fuel prices. We used it to (1) check how geopolitical risk is linked to arrivals from each country, and (2) forecast arrivals one to four months ahead with a machine-learning model, tested carefully so it never uses information that would not have been available at the time. The results are shown in a Power BI dashboard.


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## 1 · Introduction (report section 1.0)

### 1.1 Background

- **Tourism is big for Malaysia.** In 2025, tourism industries produced RM323.0 billion in value added, which is 15.9% of GDP, and employed 3.7 million people (DOSM Tourism Satellite Account).
- **International tourists matter most.** Malaysia received 25,016,698 tourists in 2024: 24.2% more than 2023, but still 4.2% below 2019. Singapore, Indonesia and China were among the biggest source countries.
- **Geopolitical risk** means the threat or occurrence of war, terrorism and tension between countries. Caldara and Iacoviello measure it with the **Geopolitical Risk (GPR) Index**, which counts newspaper articles about these events.
- **How risk can affect tourism:** people may feel less safe, governments may issue travel warnings, travellers may switch destination, air routes may be disrupted, and transport costs may rise through oil prices. Exchange rates also change how affordable Malaysia is. So the effect may differ by source country and over time.
- **Past results differ.** Lee et al. (2021) found a negative long-run link across 16 countries, but no significant link for Malaysia's tourist numbers. Papagianni et al. (2024) also found different responses across emerging economies. So an average international result cannot simply be assumed for Malaysia.

- **Our study window** runs from March 2017 to September 2026. Published arrivals end in May 2026; June to September 2026 is what we forecast.
- Because risk also moves with the economy, energy prices and the post-COVID recovery, **being able to predict with it does not prove it causes anything**. We treat risk as a forecasting signal and its estimated effects as associations only.

### 1.2 Problem statement

1. **The evidence is inconsistent,** and no study had tested the link separately for each of Malaysia's source countries instead of Malaysia as one total.
2. **There is a forecasting gap.** Malaysian research mostly looks at totals or long-run links. Short-term, country-level forecasts that combine arrivals, exchange rates, risk and energy prices, and that respect when data is really published, were not available.
3. **Decision-makers lack country-level information.** A single national figure can hide different conditions across the 20 markets. Those planning marketing, capacity and staff cannot tell whether a fall is broad or concentrated in a few countries, or see changes early enough to act.

So we evaluate, country by country, whether risk adds forecasting information once past arrivals, seasons, exchange rates and energy costs are already accounted for, and turn this into a one-to-four-month forecasting and monitoring tool that is open about its accuracy and limits.

### 1.3 Research objectives

**General objective:** evaluate the predictive relationship between geopolitical risk and international tourism demand in Malaysia, and build an understandable short-term forecasting framework for the 20 source markets, linked to SDG 8 (target 8.9) and SDG 12 (target 12.b).

1. Build a traceable monthly country dataset joining tourist arrivals, exchange rates, global and Malaysian geopolitical risk, Brent crude oil and Malaysian fuel prices (March 2017 – September 2026).
2. Assess how geopolitical risk relates to each country's arrivals after controlling for past demand, seasons, exchange rates and energy costs, treating results as non-causal associations.
3. Forecast arrivals one to four months ahead using one shared LightGBM model, and compare it with three simpler methods (naive, seasonal-naive and Ridge regression) using rolling-origin MAE and RMSE.
4. Test how robust the forecasts are under a different publication-delay assumption and during COVID, and present the results in a decision-support dashboard.

### 1.4 Scope and significance

- 20 source countries; the unit of analysis is one country in one month (a "country-month").
- Totals are for these 20 countries only. They **must not** be called Malaysia's total tourist arrivals.
- The state hotel data in our database gives background only; it is not used in the model.
- Country-level forecasts can reveal changes hidden by national totals and support monitoring, market diversification and targeted planning.
- Limits: the news-based risk index may move with other world events (especially during COVID and the recovery), results are associations, and the latest forecasts are early-warning estimates rather than exact predictions.

## 2 · Literature review (report section 2.0)

### 2.1 Geopolitical risk and tourism demand

Risk can change perceived safety, travel advisories, air routes, operating costs and destination choice, even when conflict happens far from Malaysia. Evidence is mostly negative but differs: Lee et al. (2021) found lower long-run tourism across 16 countries but no causal link for Malaysian arrivals; Papagianni et al. (2024) found negative effects in most of 14 emerging economies, stronger in China, Indonesia and Thailand. Malaysia mixes nearby land markets with long-haul air markets, so we model each country and let the risk effect differ by country.

### 2.2 Oil prices and transport costs

Transport is part of a trip's cost, and oil affects both flights and road travel. Becken and Lennox (2012) linked rising oil prices to lower tourism; Boto-Garcia (2025) found an arrival elasticity of about −0.2 across 23 European countries. But oil is an imperfect measure (airlines hedge fuel, regulated pump prices), as Seetaram (2010) and Tan and Soon (2023) note. In 2024, **41.0%** of Malaysia's tourists arrived by air and **52.4%** by land. So we use **Brent crude** for air-travel cost pressure and **Malaysian retail fuel prices** for road travel.

### 2.3 Exchange rates and affordability

Exchange rates decide how much visitors can buy in Malaysia. A weaker ringgit should make Malaysia cheaper, but Malaysian evidence shows it is not that simple: Karimi et al. (2019) found both rises and falls of the ringgit could reduce arrivals, and Tan and Soon (2023) found a weaker ringgit raised ASEAN demand but lowered demand from China, other Asian markets and Western countries. So we use each country's **own exchange rate** against the ringgit, not one overall index.

### 2.4 Malaysian tourism resilience and the SDGs

Tourism has recovered but remains vulnerable: COVID exposed weak points (Hampton et al., 2023 on Penang), and relying on a few source countries increases exposure to shocks. Country-level forecasts show whether weakness is broad or concentrated, so marketing, connectivity and staffing can be adjusted earlier. This supports the **National Tourism Policy 2020–2030**, **SDG 8.9** (sustainable tourism that creates jobs and promotes local culture) and **SDG 12.b** (tools to monitor tourism's sustainability impacts).

Forecasting does not prove sustainability by itself. The report frames the contribution in four ways:

| Dimension | Contribution |
|---|---|
| Economic | Country forecasts support steady demand and capacity planning |
| Social | Lead time for workforce preparation and protecting local cultural value |
| Environmental | Forecast peaks can be compared with carrying capacity, emissions, water, energy and waste |
| Governance | Documented sources, tested accuracy, stated limits and non-causal wording make the evidence transparent |

### 2.5 Machine learning and the research gap

- Tourism demand is seasonal, persistent and sensitive to shocks. No forecasting method wins everywhere, so each must be tested out-of-sample on the task itself.
- **Pooling**: one shared model across related series can learn from all of them (Montero-Manso and Hyndman, 2021), but differences between markets must be included (Long et al., 2019). So we add country identity and risk-by-country terms.
- **LightGBM** handles non-linear patterns and mixed inputs (Ke et al., 2017) and did well in the M5 forecasting competition (Makridakis et al., 2022), though that does not guarantee success in tourism. So it is compared with Ridge regression and simple methods.
- **SHAP** explains how a model uses its inputs, not real-world causes (Lundberg and Lee, 2017).
- **Rolling-origin testing** repeats real forecasting decisions without leakage (Tashman, 2000). MAE and RMSE keep results in tourist numbers; percentage errors become unstable for small markets (Hyndman and Koehler, 2006).
- **The gap:** Malaysian studies mostly estimate totals or long-run links, and forecasts rarely combine monthly country arrivals, bilateral exchange rates, global and Malaysian risk, energy prices, publication delays and a post-COVID stress test. Our project fills this with one shared direct model for 20 markets and 1–4 month horizons, tested against simple benchmarks using only information available at each forecast date.

## 3 · Methodology (report section 3.0)

### 3.1 Data collection and sources

| Data | Source | Used in final model? |
|---|---|---|
| Monthly tourist arrivals by nationality | Tourism Malaysia, with the Immigration Department | Yes (what we predict) |
| Exchange rates (ringgit per unit of each source currency) | Bank Negara Malaysia; a documented cross-rate with Bangladesh Bank for the taka | Yes |
| Global and Malaysia geopolitical risk | Caldara and Iacoviello | Yes |
| Brent crude oil price | World Bank | Yes |
| RON95, RON97 and diesel prices | Ministry of Finance | RON97 and Peninsular diesel |
| DOSM economic indicators (leading, coincident, lagging and diffusion indices) | DOSM | No: collected, then removed (see 3.5) |
| State hotel occupancy, guests and domestic visitors (16 states, 2017–2025) | Tourism Malaysia and DOSM | No: background only |

The 20 countries are 7 from ASEAN (Brunei, Indonesia, Myanmar, Philippines, Singapore, Thailand, Vietnam) and 13 from outside (Australia, Bangladesh, China, France, Germany, India, Japan, Nepal,

Pakistan, South Korea, Taiwan, United Kingdom, United States). A catalogue of **63 entries** records each variable's agency, dataset, link, unit, cleaning and missing values.

### 3.2 Data cleaning

- Dates, numbers and country codes were standardised; each country has one code, one currency and one region (ASEAN or not).
- **The main fix:** the raw data had **69,040 rows**, one per country per day, but arrivals and exchange rates are monthly figures copied onto every day. After checking that values never differ within a month, one row per country-month was kept: **2,300 rows** (20 countries × 115 months). Monthly values were never added up across days.
- The 80 unpublished arrival months (June–September 2026) were left empty, not filled in.
- Gaps in exchange rates were filled from official observations or the documented taka cross-rate; original and filled tables are both kept.
- Final checks found no duplicates, invalid months, negative arrivals, non-positive exchange rates or inconsistent flags.

### 3.3 Data integration

- Arrivals, exchange rates, risk and Brent keep their published monthly values. Weekly fuel prices were turned into monthly averages weighted by the days each price applied.
- Each country-month is joined to that month's national figures (risk, oil, fuel).
- Because of publication delays, the June–September 2026 forecasts use predictor values from May–August. All inputs were complete for all 80 forecast rows, and the risk and oil values used were real, not estimated.

### 3.4 Setting up the forecasting problem

- **Unit:** the country-month. Using daily rows would let different days of the same month fall into both training and testing, which would make the model look better than it is.
- **Task:** predict each country's monthly arrivals **1, 2, 3 and 4 months ahead**.
- **One shared ("pooled") model** for all 20 countries, rather than one per country, because each country has only about 111 usable months. Sharing lets countries learn from each other.
- **Direct forecasting:** the number of months ahead is given to the model as an input, and each horizon is predicted directly. The alternative ("recursive") feeds the model's own guess for next month back in, so mistakes pile up.
- **Hold-out:** the 80 unpublished country-months (June–September 2026) are the real forecast and are never used to train or test.

### 3.5 Model inputs and candidate models

The model uses **19 numeric inputs plus the country**:

| Group | Inputs |
|---|---|
| **Past arrivals** (10) | Arrivals in the latest month; arrivals 1, 2, 3, 6 and 12 months earlier; the 3-month and 6-month averages; growth versus a year earlier; arrivals in the same month last year |
| **Outside factors** (6) | Global risk, Malaysia risk, Brent oil, RON97 petrol, Peninsular diesel, exchange rate |

| | |
|---|---|
| **Calendar and structure** (3) | The target month written as two numbers on a circle (so the model knows December is next to January), and how many months ahead |
| **Country identity** | Which of the 20 countries |

**Publication-delay alignment.** Each input only uses data that would really have been published at the time of the forecast:

| Data | Delay used |
|---|---|
| Tourist arrivals | 4 months (an assumption, since no public archive of release dates was found); 3 months tested as a check |
| Global risk, Malaysia risk, Brent oil | About 1 month |
| DOSM economic indicators | Published about 2 months later (checked against DOSM's release archive); treated as 3 months to be safe |
| Exchange rates, fuel prices | Practically real-time |

The arrivals delay sets the "simulated present" at each forecast date. Example: a forecast made in September 2025 may only use arrivals up to May 2025, because June to August were not yet published.

**Input selection.** The first version included the five DOSM economic indicators. After applying their publication delay they were out of date at 3–4 months ahead and their importance fell to near zero, so they were removed. This is disclosed as a limitation.

**The four candidate models:**

| Model | How it works |
|---|---|
| Naive-last | Repeats the latest actual month for all horizons |
| Seasonal-naive | Uses the same calendar month one year earlier |
| Ridge regression | A straight-line model with a penalty that keeps its weights stable, plus a "risk × country" term for each country so each one gets its own risk effect. Mainly used to read the risk effect per country. |
| LightGBM | Many small decision trees built one after another, each correcting earlier errors. Handles numbers and categories (country) directly and captures non-linear patterns. |

### 3.6 How the models were tested

- **Rolling-origin backtest.** Five past forecast dates: March 2023, September 2023, March 2024, September 2024 and September 2025 (stable, post-recovery period, at least six months apart). At each date the model trains on earlier data, forecasts the next 1–4 months, and is compared with actual published arrivals.
- **No-leakage rule.** A training row is used only if its target month's arrivals were already published by the simulated forecast date. This is stricter and more precise than leaving an arbitrary gap.
- **Same rule for testing and final use.** Both only use rows with a full 12 months of arrivals history, so the method tested is exactly the method used for the real forecast.

- **COVID stress test.** Five more dates from March 2020 to March 2022, when arrivals from the 20 countries fell from about 2 million a month to near zero. Reported separately, never averaged with the main results.
- **Measures.** MAE (average size of the error, in tourists) and RMSE (similar, but punishes big errors more), reported for each horizon. Percentage errors were avoided because small markets (Nepal, Myanmar, Pakistan) had months near zero, which makes percentages explode.
- **Sensitivity test.** Everything was repeated with a 3-month arrivals delay to check that the choice of model doesn't depend on the 4-month assumption.

## 4 · Findings and results (report section 4.0)

### 4.1 Forecast accuracy and model choice

**Table 1: average error (MAE) in the main backtest, 4-month delay.** Lower is better.

| Months ahead | Naive-last | Seasonal-naive | Ridge | LightGBM |
|---|---|---|---|---|
| 1 | 12,752 | 23,802 | 12,117 | **11,817** |
| 2 | 9,671 | 20,404 | 9,757 | **9,571** |
| 3 | 25,157 | **21,191** | 24,258 | 21,504 |
| 4 | 21,171 | 25,413 | **20,775** | 20,983 |
| **All predictions** | 17,188 | 22,702 | 16,727 | **15,969** |

- No model is best at every horizon. LightGBM is best at 1 and 2 months; seasonal-naive is narrowly best at 3; Ridge narrowly best at 4. The gaps at 2–4 months are only 100–313 tourists (under 2%), too small to rank the models reliably with five test dates.
- LightGBM clearly beats seasonal-naive at 1, 2 and 4 months, cutting error by 50%, 53% and 17%.
- **Model choice.** Because one model makes all four horizons, the right test is total error across all predictions. LightGBM has the lowest MAE (15,969) and the lowest RMSE (36,043).
- **The choice is defensible, not decisive.** On rankings, Ridge wins 7 of the 20 date-by-horizon tests (LightGBM 4) and has a slightly better average rank (2.15 vs 2.30); LightGBM also has the larger worst single error. Ridge is a fair alternative. Picking a different model for each horizon was rejected because the gaps are small enough to be noise.

**Table 2: COVID stress test MAE, 4-month delay.**

| Months ahead | Naive-last | LightGBM |
|---|---|---|
| 1 | **9,647** | 12,805 |
| 2 | **12,366** | 16,279 |
| 3 | **15,511** | 20,730 |
| 4 | **16,134** | 20,334 |

During COVID, simply repeating the latest month was best at every horizon. No model could foresee a sudden, unprecedented shock like a border closure. The lesson: a real forecasting system should show reliability for each horizon, not one accuracy figure, so users know how far to trust it.

**Sensitivity to the arrivals delay.** With a 3-month delay, LightGBM's errors are 12,611, 10,762, 20,835 and 20,509 for horizons 1–4 (vs 11,817, 9,571, 21,504 and 20,983 with 4 months). The differences are 2–12%, and LightGBM still has the lowest total error, so the conclusion holds.

**Which inputs the model uses most** (how often the trees use each input): (1) latest month's arrivals, (2) same month last year, (3) Brent oil, (4) country, (5) RON97 petrol, (6) global risk. Recent arrivals are the strongest short-term signal, while outside factors add information at longer horizons.

### 4.2 Solution approach

**June to September 2026 forecast.** The final LightGBM, trained on all available history (7,720 training rows from 1,960 country-month starting points), forecasts for the 20 countries:

| Month | Months ahead | Forecast arrivals (20 countries) |
|---|---|---|
| June 2026 | 1 | 2,278,401 |
| July 2026 | 2 | 2,287,744 |
| August 2026 | 3 | 2,173,788 |
| September 2026 | 4 | 2,054,893 |

Stable in June and July, then a gradual decline into September, matching the usual seasonal pattern. Top five for June: Singapore 994,700, China 358,309, Indonesia 312,865, Thailand 134,150, India 108,880. These are single best estimates, without ranges.

**SDG impact pathway: forecast → decision → measured outcome.** The model estimates near-term demand; agencies can use it to adjust promotion, connectivity, staffing and readiness; those decisions should then be judged against employment, local-benefit, resource and community indicators.

| Target | What the project contributes now | Indicators needed to measure outcomes |
|---|---|---|
| SDG 8.9 | Demand planning, workforce readiness, market diversification, local cultural value | Employment, job quality, wages, skills, local SME participation, cultural benefit |
| SDG 12.b | The dashboard tracks demand, drivers, accuracy, shocks, assumptions and coverage | Energy and water per visitor, waste, emissions, capacity, congestion, resident sentiment |

The framework can be refreshed monthly with new official data, extended to more countries or destinations, and joined with SDG indicators without changing its design.

**Input completeness.** The outside inputs used by the final model (risk, Brent, fuel, exchange rates) are fully observed for every month used in the forecasts. The DOSM indicators, which would have needed estimating, are not in the final model, and the incomplete September 2026 month is never used as an input.

**Does the risk effect differ by country?** The global risk index is the same number for all 20 countries in a given month, so on its own it cannot explain differences between countries. Two methods were used:

1. **Ridge regression with "risk × country" terms.** The United States is the reference (left out so every effect can be calculated uniquely). Each country's effect is converted back to real units, and the results were double-checked by raising risk by exactly one standard deviation with everything else fixed, which reproduced every country's number.

2. **SHAP on the LightGBM model**, measuring how much risk contributes to each country's predictions.

**Ridge result:** for **16 of 20 countries**, higher risk goes with lower predicted arrivals after controlling for past arrivals, season, oil prices, exchange rates and all other inputs. **Singapore, Indonesia and China** are positive, and **Thailand** is about zero.

Change in monthly arrivals when global risk rises by one standard deviation (about 47 index points), all other inputs held fixed. The report's Table 5 shows the top 5 and bottom 6; the full 20 come from the project's results.

| Country | Change | Country | Change | Country | Change |
|---|---|---|---|---|---|
| Singapore | +38,626 | Brunei | −3,364 | Japan | −4,102 |
| Indonesia | +8,175 | Vietnam | −3,406 | Myanmar | −4,151 |
| China | +6,751 | Taiwan | −3,563 | Bangladesh | −4,300 |
| Thailand | +297 | United Kingdom | −3,580 | Pakistan | −4,447 |
| India | −563 | France | −3,767 | Germany | −4,666 |
| USA (reference) | −2,084 | Australia | −3,834 | South Korea | −4,669 |
| Philippines | −3,168 | | | Nepal | −4,807 |

**Two important cautions (from the report):**

1. **Raw vs controlled.** Across all country-months, the raw link between risk and arrivals is weakly positive (**+0.08**), but most controlled country effects are negative. The direction changes once the other inputs are included. A likely contributor is a shared time trend: risk rises over the period (its correlation with time is +0.54) while arrivals recovered after COVID. Correlation among the inputs may also play a part. These are possible explanations, not proven.
2. **Associations, not causes.** The model shows risk and arrivals moving together, not that risk causes the change. Proving cause would need a special design (such as an instrumental variable or a natural experiment), which is beyond this study.

**SHAP result.** In absolute terms, risk contributes most to big markets like Singapore and China, simply because their predictions are larger. Relative to each country's average arrivals, smaller markets rank highest: **Nepal, Myanmar and Pakistan**. These ratios are noisy for small markets, describe the model on its training data rather than proven behaviour on new periods, and show the model's internal use of risk, not real-world sensitivity.

## 5 · Dashboard (report section 5.0)

### 5.1 Overview: four pages

| Page | Purpose | Main visuals |
|---|---|---|

| | | |
|---|---|---|
| **Malaysia Tourism Overview** | Recent performance of the 20 markets | Cards: May 2026 arrivals 1.96 million (−3.1% vs May 2025); January–May 2026 arrivals 9.95 million (+0.9%). Monthly arrivals line (with December peaks). Top 5 countries for 2026 so far: Singapore 3.47M, China 1.87M, Indonesia 1.47M, Thailand 0.71M, India 0.48M. Annual arrivals and growth for complete years (2018–2024), showing the COVID fall and recovery. A note that figures cover 20 markets only. |
| **Market Analysis** | Where visitors come from | Top 10 source countries; donut of ASEAN vs non-ASEAN (ASEAN about 73% of arrivals over the whole period) |
| **Tourism Drivers** | How outside factors move with arrivals | Combination charts of arrivals vs Brent oil, exchange rate and global risk; a line of the average Malaysia risk index by year. Each chart notes that it shows co-movement, not cause. |
| **Forecast** | June–September 2026 forecast and how reliable it is | Forecast chart (gradual softening); backtest chart of actual vs predicted with an overall error figure (16.8% total error as a share of arrivals); input importance; model comparison by MAE across horizons; COVID stress test; arrivals-delay sensitivity |

### 5.2 Functions and interactivity

- **Filters (slicers):** Market Segment (ASEAN / non-ASEAN) and Source Country on the Overview; Source Country on Tourism Drivers and Forecast.
- **Cross-filtering:** clicking a bar, point or donut slice filters the other visuals on the page; clicking it again or clicking blank space clears it.
- **Tooltips:** hovering shows the month, country and value.
- **Model evaluation:** users can compare models by MAE and by horizon, and see the COVID stress test and delay sensitivity.
- For SDG use, the dashboard is the demand layer of a wider monitoring system. A future SDG page should add jobs, local enterprise, cultural benefit, emissions, resource use, waste, carrying capacity and resident outcomes.

### 5.3 The dashboard package

- Built in Microsoft Power BI Desktop (version 2.127.707.0 or later), using standard visuals only, with no add-ons. The data is stored inside the file so it opens offline.
- A static PDF of all four pages is included.
- The supporting Excel file has sheets for the country panel (2,300 rows), monthly outside factors, the 2026 forecast, backtest predictions, backtest accuracy, COVID stress test, delay sensitivity, input importance, the Ridge risk results and the SHAP risk results.
- Stated limitations: actual data runs March 2017 – May 2026; 20 markets only; June–September 2026 values are model forecasts, not DOSM figures or official forecasts; risk results are associations.

## 6 · Conclusion (report section 6.0)

- We built a monthly country dataset of arrivals, exchange rates, geopolitical risk and energy prices, and a shared LightGBM model compared with naive, seasonal-naive and Ridge benchmarks, shown in an interactive Power BI dashboard.

- LightGBM had the lowest total forecast error, though Ridge remains a reasonable alternative.
- The COVID stress test showed no model could anticipate a sudden, unprecedented shock such as a full border closure.
- The model choice stayed the same under a different publication-delay assumption.
- Risk's link with arrivals varies by country: negative for most, positive for a few (Singapore, Indonesia, China). National totals alone can miss these differences. These are associations, not causes.
- The June–September 2026 forecast is stable in June–July, then gradually lower into September, in line with seasons.
- The framework aligns most with SDG 8.9 and SDG 12.b, gives a repeatable monthly demand-monitoring layer, and can grow to more markets or destinations. It does not measure sustainability itself.
- **Future work:** prediction intervals (forecast ranges), more shock scenarios, stronger methods to establish cause and effect, and adding environmental, community, cultural and labour indicators to the dashboard.

## 7 · What is in our project folder

| Part | What it contains |
|---|---|
| **Data** | One database with six tables: the country panel (with and without filled gaps), the monthly national series (with and without filled gaps, plus flags showing which values were estimated), the state hotel panel (16 states, 2017–2025), and the 63-entry source catalogue. The same data is also in one Excel workbook.<br>A set of 20 prepared queries in three groups: **checking** the data (sources, balance across countries, filled vs original values, training vs forecast rows, country list, state data); **exploring** it (largest markets, seasonality, ASEAN vs non-ASEAN, exchange-rate swings, 2019 vs recovery, state hotel occupancy and guests); and **preparing model inputs** (lags, rolling averages, growth rates, the training table and the forecast table). |
| **Machine learning** | The full forecasting process, which runs in one step: train the final LightGBM, produce the June–September forecast and history tables, run all backtests (main, COVID and delay sensitivity), and produce the input-importance, Ridge risk and SHAP results. It also contains the saved trained model, all result tables, and a dictionary explaining every result column. Settings are fixed (including a fixed random seed) so results can be reproduced. |
| **Preliminary round** | The submitted report (PDF, 23 counted pages, submitted 22 September 2026), the dashboard package (Power BI file, PDF, Excel data, user guide), and the video presentation (8 minutes 46 seconds). Also the editable report and a preliminary slide deck. |
| **Final round** | The final slides and updated dashboard, the speaking script, and the images and videos used in the slides. |
| **Competition rules** | The official booklet and the final-round marking scheme. |

The project is published on GitHub (william20808/nexora_dosm) under the MIT licence.

## 8 · Key numbers and words

### Key numbers

| Number | Meaning | Number | Meaning |
|---|---|---|---|
| RM323.0 bn · 15.9% · 3.7M | Tourism value added, GDP share, jobs (2025) | 15,969 | LightGBM MAE, all predictions |
| 25,016,698 | Tourists to Malaysia in 2024 | 36,043 | LightGBM RMSE, all predictions |
| 20 (7 + 13) | Source countries (ASEAN + others) | 11,817 · 9,571 | LightGBM MAE at 1 and 2 months |
| 69,040 → 2,300 | Daily rows → country-months | 7 of 20 | Tests won by Ridge |
| Mar 2017 – May 2026 | Published arrivals | +0.08 | Raw risk–arrivals link |
| Jun – Sep 2026 (80) | Forecast period (rows) | 16 / 3 / 1 | Countries negative / positive / flat |
| 19 + 1 | Numeric inputs + country | +0.54 | Risk's correlation with time |
| 4 (3) | Arrivals delay, months (check) | 6th | Global risk's rank among inputs |
| 5 + 5 | Main + COVID test dates | 2,278,401 → 2,054,893 | June → September 2026 forecast |
| 7,720 | Rows used to train the final model | 41.0% / 52.4% | Tourists arriving by air / land (2024) |

### Key words

| Word | Meaning |
|---|---|
| Source market | A country tourists come from |
| Country-month | One country in one month; our unit of data |
| Geopolitical Risk (GPR) Index | A measure of world danger built from news about war, terrorism and tension |
| Horizon | How many months ahead a forecast is (1–4) |
| Forecast origin | The month from which a forecast is made |
| Publication delay | How long after a month its data is released |
| Data leakage | When a model uses information it could not have had at the time, making results look too good |
| Pooled model | One model shared by all countries |
| Direct vs recursive forecasting | Predicting each horizon directly vs feeding the model's own guesses forward |
| Rolling-origin backtest | Repeatedly pretending to forecast from past dates and checking against what happened |

| | |
|---|---|
| Stress test | Testing in an extreme period (COVID) |
| MAE / RMSE | Average error in tourists / similar but big errors count more |
| Ridge regression | A straight-line model with a penalty that keeps its weights stable |
| LightGBM | Many small decision trees built one after another, each correcting earlier errors |
| Interaction term (risk × country) | Lets each country have its own risk effect |
| Reference country | The country left out (USA) so the others can be measured against it |
| Controlling for / holding constant | Including other inputs so each effect shows only its own part |
| Standard deviation | A typical-sized change in a number (for global risk, about 47 points) |
| SHAP | A way to see how much each input contributed to a model's prediction |
| Association vs cause | Moving together vs one making the other happen |
| Point forecast | A single best estimate, without a range |